INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

LAMPS, INCANDESCENT, AIRCRAFT SERVICE,
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers incandescent lamps intended for use primarily in military aircraft service applications.

1.2 Classification. Lamps are of the following types, as specified in the applicable specification sheet.

1.2.1 Types.

a. Type I - General purpose lamps for use in applications at ambient temperatures from -67 to 185 °F (-55 to 85 °C).

b. Type II - Special purpose lamps for use in applications at ambient temperatures from -67 °F (-55 °C) and above 185 °F (85 °C).

1.3 Part or identifying number (PIN). Lamp part identifying numbers will be as specified in the applicable specification sheet. The PIN consists of the letter "M", the basic number of the applicable specification sheet (not including the revision letter), and an assigned dash number. When required for colored lamps, an additional upper case suffix letter indicating the lamp color will be used. When required for aged and selected tolerance on Mean Spherical Candela (MSCD), an additional suffix, "AS_", indicating the aged and selected tolerance will be used.

Example of part identifying number : M6363/X-1RAS15

MIL Designation	Specification Sheet Number	Dash Number	Suffix Letter for Color	Suffix for Aged & Selected
M	6363/X	-1	R	AS15

Suffix letter codes for colored lamps:

Suffix Letter	Color
R	Red
G	Green
B	Blue
Y	Yellow
W	White

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component PartsMIL-STD-202-101 - Method 101, Salt Atmosphere (Corrosion)
MIL-STD-202-103	-	Method 103, Humidity (Steady State)
MIL-STD-202-107	-	Method 107, Thermal Shock
MIL-STD-202-208	-	Method 208, Solderability
MIL-STD-202-213	-	Method 213, Shock (Specified Pulse)
MIL-STD-202-214	-	Method 214, Random Vibration
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials

(See ASSIST database for list of specification sheets)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C81.61	-	Electric Lamp Bases – Specifications for Bases (Caps) for Electric Lamps

(Copies of this document are available online at https://webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM D6123/D6123M	-	Standard Specification for Pressure-Sensitive Tape for Light-Duty Packaging and General Purpose Masking

(Copies of this document are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS4156	-	Color-Coded Incandescent Flange Base T1 and T1-3/4 Lamps for Voltage Identification
SAE AS25050	-	Colors, Aeronautical Lights and Lighting Equipment

(Copies of these documents are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual lamp requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Lamps furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.4 and 6.3).

3.3 Selection of parts, materials, and finishes. Selection of parts, materials, and finishes that are not specifically designated herein or in the applicable specification sheet shall meet or exceed the requirements of this specification. Acceptance or approval of any constituent part, material, or finish shall not be construed as a guarantee of the acceptance of the finished product.

3.4 Materials. The materials shall be as specified herein. However, when a definitive material is not specified, a material shall be used that will enable the lamps to meet the requirements of this specification and applicable specification sheets.

3.4.1 Metals. Metals shall be corrosion resistant or treated to resist corrosion caused by fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal aircraft service. Lamp base and contact materials shall be resistant to electrolytic corrosion when installed in metal sockets and holders.

3.4.1.1 Dissimilar conductive materials. Dissimilar conductive materials, as defined in MIL-STD-889, shall not be used in intimate contact with each other unless protected against electrolytic corrosion.

3.4.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.3 Solder. The solder type selected shall meet the intended use of the lamp or its application, and temperature rating as defined by the applicable specification sheet.

3.5 Design and construction. Lamps shall be of the design, color, and physical dimensions shown in the applicable specification sheet (see 6.2).

3.5.1 Type I lamps. Type I lamps shall be clear, colored, or reflective coated incandescent lamps for use at ambient temperatures up to 185 °F (85 °C).

3.5.2 Type II lamps. Type II lamps shall be clear, colored, or reflective coated incandescent lamps for use at ambient temperatures above 185 °F (85 °C), or as otherwise specified in the applicable specification sheet.

3.5.3 Reflector lamps. Reflector lamps may be fabricated by applying an external reflective coating to a specially shaped globe, or by placing the filament between an internally coated reflector and a cover glass that is permanently sealed to the reflector. In the latter arrangement, the use of an internal lamp within the globe is optional (see 6.5.8).

3.5.4 Coatings. The exterior coating on lamps (both reflective and colored) shall be such that it will withstand the rigors of normal handling, service use, and storage.

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3.5.5 Colored lamps. The lamps shall be uniformly colored with no perforations or opening through which non-colored light can be emitted. All light emitted from colored lamps shall conform to SAE AS25050 for the color specified. Color may be obtained by the use of either colored glass or a coating on the glass.

3.5.6 Internal wiring. Provision shall be made to prevent any type of electrical shorting inside the metal base of lamps.

3.5.7 Filament. The structure and processing of the filament material shall be of a quality to meet the requirements specified herein and in the applicable specification sheet. The arrangement of the filament and filament support shall be as designated by the applicable specification sheet. When not specified by the specification sheet, the filament arrangement and support shall be at the manufacturer's option, to meet the requirements specified herein and in the applicable specification sheet.

3.5.8 Lamp bases. Where applicable, lamp base configuration and dimensions shall conform to the base type designation of ANSI C81.61 as specified in the specification sheet. Where a lamp base is required that is not covered by ANSI C81.61, the configuration and dimensions shall conform to those specified in the specification sheet.

3.5.9 Optional design. Design details not specified or specifically dimensioned are optional within the maximum design envelope specified for the applicable lamp configuration. Minor variations will be permitted provided the requirements are met and the lamp is totally interchangeable with lamps that use the identified design details. Any changes in design that affect form, fit, or function of the lamp require written approval by NAVSEA.

3.6 Performance. The applicable requirements are defined by the applicable specification sheet if not defined herein.

3.6.1 Ratings. When operating at rated voltage, lamps shall conform to the current, power, and light ratings specified in the applicable specification sheet (see 4.6.3 and 6.4).

3.6.1.1 Light output.

3.6.1.1.1 Initial candlepower. Lamps furnished under this specification shall have an initial candlepower output within the range specified in the applicable specification sheet when tested as specified (see 4.6.3).

3.6.1.1.2 Candlepower maintenance. Unless otherwise specified in the specification sheet, the average candlepower output of sample group lamps shall not fall below 80 percent of the initial candlepower output at 70 percent of average rated life. The exterior coating of lamps (if so furnished, both reflective and colored) shall not be damaged or discolor, crack, fade, blister, or peel (see 4.6.5, 6.5.2, 6.5.3, 6.5.4, 6.5.6, 6.5.7, and 6.5.12).

3.6.2 Life. Life requirements for lamps furnished under this specification shall be average rated laboratory life, as specified in the specification sheet. The sample group lamps shall operate for the average rated life shown in the specification sheet for the specific lamp part number. The average rated life shall be based on the average life of groups of lamps operated at rated voltage under controlled laboratory conditions when tested as specified (see 4.6.4, 4.6.4.1, and 6.4).

3.6.3 Coating adhesion. When tested for adherence to the glass bulb, the exterior coatings (both reflective and colored), shall be such that they shall not be damaged or blister, crack, or peel from the globe when tested as specified (see 4.6.6 and 4.6.8).

3.6.4 Solderability. Unbased lamps having tinned lead terminals shall have the leads tested for solderability. The solder dipped surface of the terminal shall be at least 95 percent covered with a new, smooth solder coating. The remaining 5 percent of the terminal surface may contain small pinholes, voids, or rough spots not concentrated in any one area. Bare base metal and areas where the new solder dip failed to cover the original coating are indications of poor solderability. The solderability test shall be performed as specified (see 4.6.14).

3.6.5 Environmental. Unless otherwise specified in the specification sheet, lamps shall perform within the specified rating limits when subjected to the following environmental requirements for the following specified times, temperatures, and environmental conditions:

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3.6.5.1 High temperature. When operating at rate voltage, the lamps shall perform to a minimum of 70 percent of the average rated life specified in the applicable specification sheet (up to 1000 hours maximum test time) while exposed to the specified maximum temperature for the type lamp being tested. Lamps shall show no evidence of burn-out or damage to any lamp component as a result of exposure to temperature extremes. Exterior coated lamps (both reflective and colored) shall show no evidence of damage, discoloration, cracking, fading, blistering, or peeling of the coating (see 4.6.8).

3.6.5.2 Thermal shock. When operating at rated voltage, the lamps shall perform when subjected to severe changes in thermal conditions. Lamps shall show no evidence of burn-out or damage to any lamp component as a result of exposure to temperature extremes. Exterior coated lamps (both reflective and colored) shall show no evidence of damage, discoloration, cracking, fading, blistering, or peeling of the coating (see 4.6.9).

3.6.5.3 Random vibration. When operating at rated voltage, the lamps shall perform when subjected to random vibrations in a frequency range from 5 to 2000 hertz. Lamps shall show no evidence of burn-out, damage to any lamp component, or loose parts (see 4.6.10).

3.6.5.4 Shock. When operating at rated voltage, the lamps shall perform when subjected to a shock test using peak "g" load values as specified in the applicable specification sheet. If no "g" load value is specified, the shock test shall be performed using a minimum of 30 "g's" peak load. Lamps shall show no evidence of burn-out, damage to any lamp component, or loose parts (see 4.6.11)

3.6.5.5 Humidity. After extended, non-operating exposure to hot, humid atmosphere, the lamps shall exhibit globe to base integrity and shall exhibit no evidence of atmospheric leakage into the lamp. Following the humidity test, lamps shall conform to the ratings (current, power, and light output) specified in the applicable specification sheet when operated at rated voltage (see 4.6.12 and 6.4).

3.6.5.6 Salt spray. After extended, non-operating exposure to a hot, humid, salt atmosphere, the lamps shall exhibit no signs of corrosion. Following the salt spray test, lamps shall conform to the ratings (current, power, and light output) specified in the applicable specification sheet when operated at rated voltage (see 4.6.13 and 6.4).

3.7 Identification of product. Unless otherwise specified (see 6.2), all lamps shall be marked for identification in accordance with the marking specified in the specification sheet. Markings shall be clear, legible, and durable (see 4.6.1).

3.7.1 Color coding for voltage identification. When specified in the specification sheet, lamp base insulators shall be color coded in accordance with SAE AS4156 in order to identify voltage ratings of subminiature lamps.

3.8 Examination of product. Lamps shall conform to dimensional requirements specified in the applicable specification sheet and the workmanship and marking requirements herein and in the applicable specification sheet (see 4.6.1).

3.9 Workmanship. Each lamp, including all parts and accessories, shall be fabricated and finished, free of blemishes and defects that will adversely affect its life, form, fit, or function. Soldering, welding, brazing, cementing, and wiring shall be thorough, and alignment of parts shall be accurate. Each lamp shall be thoroughly cleaned. Loose, spattered, or excess solder, metal chips, flux, and other foreign material shall be removed (see 4.6.1, 6.5.1, and 6.5.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Conformance inspection (see 4.5).

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4.2 Inspections and tests. The inspections and tests specified herein are intended to verify that the items produced meet or exceed the requirements specified.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202.

4.3.1 Light output criteria. Unless otherwise specified (see 6.2), light output criteria shall be as specified in the applicable specification sheet. The test procedure for determining the change in light output before and after burning shall be the same for any one type lamp.

4.3.2 Lamp operation. Unless otherwise specified (see 6.2), lamps shall be operated at the rated voltage specified in the applicable specification sheet. Unless otherwise specified in the specification sheet, the voltage tolerance shall be within ±0.5 percent of the rated voltage.

4.3.3 Lamp mounting. Unless otherwise specified (see 6.2), lamps shall be tested in a lampholder or fixture for the lamp under test. The lampholder or fixture shall be rigidly mounted with no special provisions for absorbing or isolating the effects of shock, vibration, or temperature. When testing directional beam lamps that require light distribution measurements to be made relative to a specified reflector for flight axis, provision shall be made in the test fixture for rotation of the mounted lamps to provide the required measurement axis. Care shall be taken to ensure that the light output measurements for each directional beam lamp is made on the same mechanical angle each time so that changes in the beam axis can be detected.

4.3.4 Precautions. The following precautions shall be observed when performing the high temperature tests:

a. The test chamber shall be capable of maintaining the specified temperature with ±10 °F(±5 °C). The test chamber shall be operated for 1 hour prior to installation of lamps to ensure that the temperature has stabilized.

b. Temperature measurements shall be made with thermocouples having leads at least 20 inches long within the oven. This will minimize the conduction of heat away from the junction.

c. Baffles shall be placed within the oven to shield the lamps and thermocouple junctions from direct radiation from the heating elements and surfaces of the oven.

d. The lamps shall be mounted so that there is a clear space between the lamps equal to at least the width of one lamp. The lamps shall be mounted so that the flux from any lamp does not directly strike any other lamp.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the qualifying activity (see 6.3) on sample lamps produced with equipment and procedures normally used in production. The samples may be selected at random from a recent production, or, if lamps of a recent production are not available, the sample may be fabricated. The qualification inspection requirement specified herein is classified as follows:

a. Full qualification – Lamp samples shall be subjected to inspections and tests specified in table I.

b. Qualification by similarity (see 4.4.1) – Lamp samples subjected to inspections and tests specified in table II.

4.4.1 Extension of qualification by similarity. Unless otherwise specified by the qualifying activity, qualification of similar lamps from the same or different specification sheets may be extended from a previous fully qualified lamp.

4.4.1.1 Conditions for qualification by similarity. Additional lamps submitted for qualification by similarity shall be products manufactured at the same manufacturing plant location, under the same manufacturing processes and quality controls as the fully qualified primary lamp(s). Unless otherwise approved by the qualifying activity, additional lamps must be of the same generic design and construction as the qualified primary lamp(s).

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TABLE I. Qualification inspection.

Inspection/Test	Number of Sample Units	Requirement	Test Method
All sample groups			
Examination of product	All	3.8	4.6.1
Burn-in/seasoning		---	4.6.2
Sample group 1			
Ratings (current, power, initial light output)	10	3.6.1	4.6.3
Life 1/		3.6.2	4.6.4
Light output maintenance 2/		3.6.1.1.2	4.6.5
Coating adhesion*		3.6.2	4.6.6
Solderability*		3.6.4	4.6.14
*Where applicable			
Sample group 2			
High temperature	3	3.6.5.1	4.6.8
Sample group 3			
Thermal shock	3	3.6.5.2	4.6.9
Sample group 4			
Random vibration	3	3.6.5.3	4.6.10
Sample group 5			
Shock	3	3.6.5.4	4.6.11
Humidity		3.6.5.5	4.6.12
Salt Spray		3.6.5.6	4.6.13
Sample group 6			
Spares only to replace unrelated failures (see 6.5.13)	3	---	---

NOTES:

1/ For average rated life specified in the applicable specification sheet up to 1000 hours maximum. For longer life lamps, the qualifying activity reserves the right to satisfy the life requirement by verification and acceptance of manufacturer's data from in-process conformance tests.

2/ Unless otherwise specified (see 6.2), for 70 percent of average rated life specified in the applicable specification sheet up to 1000 hours maximum life.

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TABLE II. Qualification inspection by similarity.

Inspection/Test	Number of Sample Units	Requirement	Test Method
Burn-in/seasoning	---	---	4.6.2
Ratings (current, power, initial candlepower)	3	3.6.1	4.6.3
Random vibration	---	3.6.5.3	4.6.10
Colored lamps (color, high temp, coating adhesion)	---	3.5.5	4.6.7
	---	3.6.5.1	4.6.8
	---	3.6.3	4.6.6
Solderability (where applicable)	3 (spares only)	3.6.4	4.6.14

4.4.2 Sample for full qualification. Unless otherwise specified by the qualifying activity, the test sample shall consist of 25 lamps of each specified part number for which full qualification is desired. The 25 lamps shall be divided into six sample groups: one sample group of 10 lamps (Sample Group 1), and 5 samples groups of 3 lamps each (sample groups 2 through 6). Sample groups 1 through 5 shall be subjected to and must pass the test series in table I for the group number indicated. The sixth group of samples shall be used for spares to replace unrelated failures (see 6.5).

4.4.2.1 Samples for qualification by similarity. Unless otherwise specified by the qualifying activity, the test sample shall consist of 10 lamps of each specific part number for which qualification by similarity is desired. For purposes of qualification testing, 3 lamps of each part number shall be subjected to and must pass the test specified in table II. Three lamps of each part number will be used as spares to replace unrelated failures (see 6.5).

4.4.3 Failure. Failure to meet the requirements specified herein shall be cause for disapproval or removal of product qualification.

4.5 Conformance inspection.

4.5.1 Inspection lot. An inspection lot shall consist of all lamps of one designation and size, from an identifiable production period, from one manufacturer, and one plant location. When production lots are fractionalized each partial quantity shall be considered a separate inspection lot.

4.5.2 Inspection of product.

4.5.2.1 Group A and B inspection. Inspection of product shall include group A and B inspections (see table III). Random sampling for group A and B inspections shall be accomplished throughout the complete production lot manufacturing process in accordance with the sampling plan and acceptance level as specified (see 6.2). In the event of failure of sample lamps to pass group A and B inspections, the inspection lot shall be rejected.

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TABLE III. Conformance inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Group A		
Examination of product	3.8	4.6.1
Group B		
Ratings (current, power, initial light output)	3.6.1	4.6.3
Solderability (where applicable)	3.6.4	4.6.14
Colored lamps	3.5.5	4.6.7
Group C		
Life	3.6.2	4.6.4
Light output maintenance	3.6.1.1	4.6.5
Coating adhesion (where applicable)	3.6.3	4.6.6

4.5.3 Periodic in-process inspection. Periodic in-process inspection shall consist of group C inspection.

4.5.3.1 Group C inspection. Sample lamps selected from the manufacturer's normal in-process inspection procedure shall be subjected to the group C inspections specified in table III. The manufacturer's normal sampling, based upon production quantity or time periods, shall be used, and samples shall have passed the applicable group A and B inspections. Group C inspection samples shall be representative of production.

4.5.3.1.1 Disposition of samples. Sample lamps that have been subjected to group C inspection shall not be delivered to the Government on contract or purchase order.

4.6 Test methods and examination.

4.6.1 Examination of the product. The sample group lamps shall be visually and dimensionally examined for conformance to this specification and the applicable specification sheet.

4.6.2 Burn-in/seasoning. Prior to the start of testing, the sample group lamps shall be seasoned by being operated at the rated voltage for a period of time equal to 1percent of the rated life specified in the applicable specification sheet or a maximum of 10 hours.

4.6.3 Ratings. The sample group lamps shall be mounted on an open rack, locked in a vibration free position, and connected to a power source and required instrumentation.

4.6.3.1 Photometric tests. Photometric tests shall be performed using standard lamps traceable to National Institute of Standard and Technology. Lamps standards shall be calibrated on a yearly basis minimum.

4.6.3.2 Initial ratings. After burn-in/seasoning, the sample group lamps shall be energized at rated voltage, and the current, power, and initial light output shall be measured, recorded, and compared to rating specified in the applicable specification sheet.

4.6.3.3 Post test ratings. When required by the specified test procedure, the sample group lamps shall be energized at rated voltage, and the current, power, and light output shall be measured, recorded, and compared to ratings specified in the applicable specification sheet.

4.6.4 Life. The sample group lamps shall be mounted on an open rack, locked in a vibration-free position, and connected to a suitable power source and required instrumentation. The lamps shall be operated at rated voltage to determine that average lamp life conforms to that specified in the applicable specification sheet. For qualification purposes, the life test shall be performed up to 1000 hours maximum.

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4.6.4.1 Life test average determination. The average laboratory lamp life is determined by operation of the lamps at rated voltage. The average represents the number of hours that it would take for 50 percent of the test samples to fail. Lamps are considered as having failed when they burn out or light output decreases by more than 20 percent of initial reading. For qualification purposes, when test samples are exceeding the specification requirement prior to the point of failure of 50 percent of the sample lamps, the life test may be terminated prior to 50 percent sample failure.

4.6.5 Candlepower maintenance. The sample group lamps shall be tested to determine that at 70 percent of the specified average rated life, or as specified in the applicable specification sheet, the light output does not fall below 80 percent of the initial light output. For colored lamps, all light emitted shall conform to SAE AS25050 for the color specified in the applicable specification sheet.

4.6.6 Coating adhesion. The sample group lamps shall be tested to determine the coating adhesion by tape stripping using the following procedure. A minimum of 2 inches or a longer piece of pressure sensitive adhesive tape, 0.75 inches in width, conforming to ASTM D6123/D6123M, type I, shall be pressed firmly onto a flat or cylindrical coated surface of the sample lamp or as specified in the applicable specification sheet, rubbing out all air bubbles under the tape. Ten seconds shall be allowed for the test area to return to room temperature. Then, grasping a free end of the tape, and in a rapid motion, the tape shall be pulled back upon itself at 180 degrees (in such a manner that the tape is folded back-to-back during the procedure). This test shall be performed both before and after the sample group lamps are operated for 10 percent of the rated life. After testing, the sample group lamps shall be checked for conformance to 3.6.3. Any sample lamp indicating failure shall be cause for rejection of the sample group. When performed as part of the group C tests, failure of the sample group to meet the coating adhesion requirements shall be cause for rejection.

4.6.7 Colored lamps. Colored lamps shall be operated at rated voltage and the light emitted shall be measured for conformance to SAE AS25050 for the specified color.

4.6.8 High temperature. The sample group lamps shall be mounted on an open rack, locked in a vibration-free position, and operated at rated voltage for a minimum of 70 percent of average rated life or as specified in the applicable specification sheet up to a maximum of 1000 hours for qualification testing. Lamps designated as type I shall be operated at 185 °F(85 °C) and lamps designated as type II shall be operated at the maximum temperature specified in the applicable specification sheet. The time of any individual sample lamp failure and reason for failure shall be recorded. After test, lamps shall be checked for conformance to 3.6.5.1.

4.6.9 Thermal shock. The sample group lamps shall be subjected to a thermal shock test as specified in MIL-STD-202-107, with the exceptions indicated in table IV. This test shall be performed with the lamps operating at rated voltage. Table IV shall be substituted for the thermal shock test conditions table specified in MIL-STD-202-107. Continue steps 1 through 4 of table IV for 25 cycles. The time of any individual sample lamp failure and reason for failure shall be recorded. After testing, lamps shall be checked for conformance to 3.6.5.2.

TABLE IV. Thermal shock test conditions.

	Type I Lamps			**Type II Lamps**		
Step	Temperature		Time per cycle	Temperature		Time per cycle
	°C	°F		°C	°F	
1	-55+1	-67+2	See 2/	-55+1	-67+2	Table 107-2
2	25+2	77+5	5 min max	25+2	77+5	5 min max
3	85+2	185+5	Table 107-2	See 1/	See 1/	Table 107-2
4	25+2	77+5	5 min max	25+2	77+5	5 min max

NOTE:

1/ High temperature as specified in the applicable specification sheet.

2/ Time per cycle shall meet the minimum exposure time in air at temperature extremes specified in MIL-STD-202-107.

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4.6.10 Random vibration. The sample group lamps shall be subjected to a random vibration spectrum between 5 and 2000 hertz with power spectral densities as specified in the applicable specification sheet for each lamp type.

4.6.10.1 Vibration system control analysis. The output of the vibration machine shall be presented graphically as power-spectral density versus frequency. The spectral density values shall be as specified in the applicable specification sheet. The vibration apparatus control and analysis of vibration shall be in accordance with MIL-STD-202-214.

4.6.10.2 Test setup applicable to all lamps. The sample group lamps shall be vibrated for a total of 30 minutes; 7.5 minutes with the lamps nonoperating and 22.5 minutes with the lamps operating at rated voltage. Nonoperating vibration shall be in "Z" axis as indicated in the applicable specification sheet. During the operating mode, the lamps shall be vibrated for 7.5 minutes in each of 3 mutually perpendicular axes as defined in the applicable specification sheet.

The vibration with lamps operating shall start immediately after lamps are lit without any additional burn-in. Either the fixture of the lamps may be rotated to achieve the positioning of lamps in each of the axes. The time to failure of any individual sample lamp and the observed reason for failure shall be reported. After this test, the sample group lamps shall be inspected for conformance to 3.6.5.3.

4.6.11 Shock. The sample group lamps shall be subjected to a shock test in accordance with test condition J of MIL-STD-202-213, with the exceptions indicated herein. This test shall be performed with the lamps operating at the rated voltage. A peak "g" load value as specified in the applicable specification sheet shall be substituted for that specified in test condition J of MIL-STD-202-213. If no "g" load value is specified in the specification sheet, the shock test shall be conducted using a 30 g's peak load. The time of failure of any individual sample lamp and reason for failure shall be recorded. After this test, the sample group lamps shall be inspected for conformance to 3.6.5.4.

4.6.12 Humidity. The sample group lamps, nonoperating, shall be subjected to a humidity test in accordance with test condition A (240 hours) of MIL-STD-202-103 and visually inspected to determine conformance with 3.6.5.5. After this test, the sample group lamps shall be operated to determine conformance to 3.6.5.5.

4.6.13 Salt spray. The sample group lamps nonoperating, shall be subjected to a salt spray test in accordance with test condition B (48 hours) of MIL-STD-202-101 and visually inspected to determine conformance with 3.6.5.6. After this test, the sample group lamps shall be operated to determine conformance to 3.6.5.6.

4.6.14 Solderability. Sample group unbased lamps having wire lead terminations shall be tested in accordance with MIL-STD-202-208 and evaluated for solderability of wire leads as specified in 3.6.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lamps are intended for use primarily in military aircraft lighting applications where equipment and aircraft must operate under severe performance, storage, and environmental conditions. Typical lamp usages include applications in instruments and panel illumination, control and signal indicators, anti-collision lights, wing position lights, taxiing, and landing lights. Lamps procured to this specification are form, fit, and function replacements for prior revisions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. The applicable specification sheet.

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c. The applicable QPL.

d. Part identifying number of lamp required (see applicable specification sheet and 1.3).

e. Color of lamp required (see applicable specification sheet and 1.3).

f. Lamp identification markings (see applicable specification sheet and 3.7).

g. Inspection conditions (see 4.3).

h. Light output criteria (see applicable specification sheet and 4.3.1).

i. Lamp operation voltage (see applicable specification sheet and 4.3.2).

j. Lamp mounting (see applicable specification sheet and 4.3.3)

k. Light output maintenance test/inspection criteria (see applicable specification sheet and table I).

l. Sampling plans and acceptance levels for group A and group B inspections (see 4.5.2.1).

m. Packaging requirements (see 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 6363 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Rated life. The specified average rated life (see 3.6.2) is based upon operating the lamps at the specified design voltage. Where performance of life testing at rated voltage is considered to be impractical, the use of accelerated life testing based on the 12th power rule is permissible.

$$L_A=L_R(V_R/V_O)^{12}$$

Where L_A = Accelerated life

L_R = Rated life at design voltage

V_R = Rated voltage

V_O = Operating voltage

Upon application to, and approval of the acquiring activity, the manufacturer may provide test results from their in-process conformance testing to substantiate the life requirements.

6.5 Definitions. For the purposes of defining various types of defects or terminology that are referred to in this specification or in the applicable specification sheet, the following definitions apply:

6.5.1 Blemish. A visual nonfunctional imperfection of the symmetry of the lamp globe or base.

6.5.2 Blister. A visible separation of a coating from the glass bulb that forms an upraised, unbroken bubble.

6.5.3 Crack. A narrow lengthwise opening, split, or fissure produce by a partial separation of the coating from the bulb surface.

6.5.4 Damage. Any lamp component defect that prohibits the lamp from conforming to its basic functional and light distribution requirements.

6.5.5 Defect. A fault in workmanship or manufacture that affects the proper functioning of the item.

6.5.6 Discolor. A distinct change in the hue of color or coating that adversely affects the lamp's functional color or reflective qualities and reduces its illumination characteristics to an out of specification condition.

6.5.7 Fade. A loss of brightness or vividness of color such as to adversely affect the lamp's functional color or reflective qualities and reduces its illumination characteristics to an out of specification condition.

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6.5.8 Filament. The types of filaments are as follows:

C = single helical coil

CC = coiled coil or double helical coil

F = flat coil

S = straight uncoiled wire

6.5.9 Light center for C-2 shaped filaments. Light center is a point lying in a plane through the axis of the filament coils midway between the "top" and "bottom" of the lighted filament and midway between the outer edges of the filament coils at the horizontal level of the light center.

NOTE: Filament top is the upper extremity of the highest lighted turn (away from base) in either coil segment. Filament bottom is the underside of the lowest lighted turn (toward base) in either coil segment. The lowest lighted turn is considered for this purpose as being the first turn above the first inside angle between turns of the projected filament image.

6.5.10 Light center for C-6 shaped filaments. Light center is a point lying in the axis of the filament coil midway between the end turns of the lighted filament.

6.5.11 Light center length. Light center length is the distance from light center to the specified reference point on the lamp.

a. Reference points on the lamp base for light center length measurement:

(1) Bayonet base – Top of base pins.

(2) Single contact (SC) miniature flanged – Top of bosses in the flange.

(3) SC index – Top of base pin nearer bottom contact.

(4) SC midget flange – Top of flange.

(5) Submidget flange – Top of flange.

6.5.12 Peel. Defined as any one of the following conditions for:

a. Colored lamps:

(1) No more than one area having a coating void not to exceed 0.0625 inch average diameter.

(2) No more than two areas having coating voids not to exceed 0.0312 inch average diameter or combined total of 0.0625 inch diameter.

(3) No more than five areas having coating voids of less than 0.0312 inch average diameter or a combined total of 0.0625 inch diameter.

b. Reflector lamps:

(1) With the lamp operating at rated voltage no more than four non-adjacent voids should be visible when the lamp is checked from a distance no closer than 18 inches. Void diameter should not exceed 0.050 inch. These requirements do not apply within 0.050 inch of the glass to base junction and the coating to glass junction.

6.5.13 Unrelated failure. A lamp failure, malfunction, or breakage, occurring during testing, which is not attributable to the effect of the test procedure. Examples:

a. Lamp inadvertently damaged or broken during handling or fixturing.

b. Malfunction of test equipment causing lamp under test to fail or malfunction.

6.6 Subject term (key word) listing.

Aircraft lamps
Aircraft lighting
Bulbs
Illumination
Lighting
Lights

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 71

Review activities:
Navy – AS
DLA – GS

Preparing activity:
Navy – SH
(Project 6240-2020-020)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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